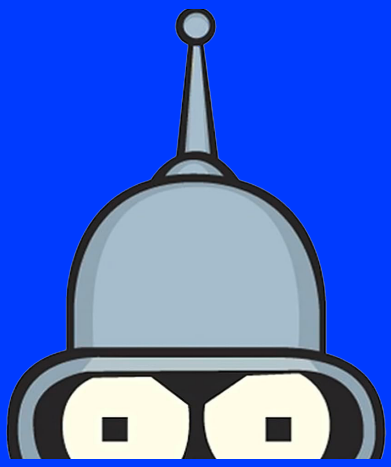
{EPITECH}

COREWAR - INSTRUCTIONS_

< ELEMENTARY PROGRAMMING IN C />



COREWAR - INSTRUCTIONS

The instructions

Champions have the ability of executing any of 16 instructions (detailed and established in the `op.c` source file provided). Below you will find a list explaining their parameters and behavior.
Don't worry if you don't yet understand everything written there; a lot of this information will actually come in handy during the Corewar project, not necessarily now!

In the dump examples given for the instructions below, the color code is as such :

```
Terminal                                                    - + x
  ✓ Green: The current program counter location.
  ✓ Magenta: The champion's byte code.
  ✓ Cyan: What changed following the previous instruction(s).
```

For readability purposes, most of the examples below are shown as if there was only one champion running in the Corewar virtual machine.

Examples are not always illustrated in terminal boxes.
Terminal boxes do not always represent examples above them.

Take some time, read the byte code, and understand what the champion is doing. It is tedious, but necessary !

0x01 (live)

Parameters: `direct` `player_id`

Cycles: 10

Behavior:
Indicates that the player whose identifier equals `player_id` is alive.

0x02 (ld)

Parameters: `direct/indirect source, register destination`

Cycles: 5

Behavior:
Loads the value of the `source` into `destination`.

This operation sets the carry to 1 if `destination` is zero after the operation, otherwise 0.

Example:
`ld %34, r3` loads the number 34 into `r3`.

```
Terminal
Cycle: 0
Registers:
  ld(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000007    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 02 90 00 00 00 22 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 5
Registers:
  ld(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000022    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000007    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 02 90 00 00 00 22 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

`ld 34, r3` loads `REG_SIZE` bytes starting at the address `PC + 34 % IDX_MOD` into `r3`.

0x03 (st)

Parameters: `register source, indirect/register destination`

Cycles: 5

Behavior:
Stores the value of `source` into `destination`.

Example:
`st r1, 32` stores the content of `r1` at the address `PC + 32 % IDX_MOD`.

```
Terminal
Cycle: 0
Registers:
  st(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000000    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 03 70 01 00 20 03 50 01 07 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 5
Registers:
  st(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000005    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 03 70 01 00 20 03 50 01 07 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000020: 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 10
Registers:
  st(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000001    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000005    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 03 70 01 00 20 03 50 01 07 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000020: 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

`st r1, r7` copies the content of `r1` into `r7`.

0x04 (add)

Parameters: `register a, register b, register destination`

Cycles: 10

Behavior:
Adds the value of `a` to `b` and stores the result in `destination`

This operation sets the carry to 1 if `destination` is zero after the operation, otherwise 0.

Example:
`add r1, r2, r3` adds the content of `r1` and `r2` and puts the result into `r3`.

```
Terminal
Cycle: 5
Registers:
  ld_add(1): alive
    r1 : 00000001    r2 : 0000002A    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000007    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 02 90 00 00 00 2A 02 04 54 01 02 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 15
Registers:
  ld_add(1): alive
    r1 : 00000001    r2 : 0000002A    r3 : 0000002B    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 0000000C    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 02 90 00 00 00 2A 02 04 54 01 02 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

0x05 (sub)

Parameters: `register a, register b, register destination`

Cycles: 10

Behavior:
Subtract the value of `b` from `a` and stores the result in `destination`.

This operation sets the carry to 1 if `destination` would be negative, otherwise 0.

Example:
`sub r1, r2, r3` subtract the content of `r2` from `r1` and puts the result into `r3`.

```
Terminal
Cycle: 5
Registers:
  st_sub(1): alive
    r1 : 00000001    r2 : 0000002A    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000007    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 02 90 00 00 00 2A 02 05 54 01 02 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 15
Registers:
  st_sub(1): alive
    r1 : 00000001    r2 : 0000002A    r3 : FFFFFFD7    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 0000000C    carry: 1
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 02 90 00 00 00 2A 02 05 54 01 02 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

0x06 (and)

Parameters: `direct/indirect/register a, direct/indirect/register b, register destination`

Cycles: 6

Behavior:
Performs a binary AND between `a` and `b` and stores the result into the `destination`.

This operation sets the carry to 1 if `destination` is zero after the operation, otherwise 0.

Example:
`and 1, %42, r2` puts `*(PC + 1) & 42` into `r2`.

```
Terminal
Cycle: 0
Registers:
  and(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000000    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 06 E4 00 01 00 00 00 2A 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 6
Registers:
  and(1): alive
    r1 : 00000001    r2 : 00000020    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000009    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 06 E4 00 01 00 00 00 2A 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

`and r2, %0, r3` puts `r2 & 0` into `r3` and sets the carry to 1.

0x07 (or)

Parameters: `direct/indirect/register a`, `direct/indirect/register b`, register `destination`

Cycles: 6

Behavior:
Performs a binary OR between `a` and `b` and stores the result into the `destination`.

This operation sets the carry to 1 if `destination` is zero after the operation, otherwise 0.

Example:
`or 1, %42, r2` puts `*(PC + 1) | 42` into `r2`.

```
Terminal
Cycle: 0
Registers:
  or(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000000    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 07 E4 00 01 00 00 00 2A 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 6
Registers:
  or(1): alive
    r1 : 00000001    r2 : 000000BE    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000009    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 07 E4 00 01 00 00 00 2A 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

`or r2, %0, r3` puts `r2 | 0` into `r3`.

0x08 (xor)

Parameters: `direct/indirect/register a`, `direct/indirect/register b`, register `destination`

Cycles: 6

Behavior:
Performs a binary XOR between `a` and `b` and stores the result into the `destination`.

This operation sets the carry to 1 if `destination` is zero after the operation, otherwise 0.

Example:
`xor 1, %42, r2` puts `*(PC + 1) ⊕ 42` into `r2`.

```
Terminal
Cycle: 0
Registers:
  xor(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000000    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 08 E4 00 01 00 00 00 2A 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 6
Registers:
  xor(1): alive
    r1 : 00000001    r2 : 0000009E    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000009    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 08 E4 00 01 00 00 00 2A 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

`xor r2, %0, r3` puts `r2 ⊕ 0` into `r3`.

0x09 (zjmp)

Parameters: `index` `offset`

Cycles: 20

Behavior:
Perform a jump (move the PC) by `offset` bytes if the carry is worth 1. Otherwise, does nothing but consumes the same time.

This specific instruction **does not** have a coding byte !

It **always** takes a direct index (eg: a direct on `IND_SIZE` bytes).

Example:
`zjmp %25` jumps forward by 25 bytes (`PC = PC + 25`).

```
Terminal
Cycle: 5
Registers:
  ld_zjmp(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000007    carry: 1
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 02 90 00 00 00 00 02 09 00 19 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 25
Registers:
  ld_zjmp(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000020    carry: 1
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 02 90 00 00 00 00 02 09 00 19 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

0x0a (ldi)

Parameters: `index/register offset_a, index/register offset_b, register destination`

Cycles: 25

Behavior:
Adds together `offset_a` and `offset_b` (let's call that SUM) and loads the value at `PC + SUM % IDX_MOD` into `destination`.
This operation sets the carry to 1 if `destination` is zero after the operation, otherwise 0.

Example:
`ldi %1, %2, r3` reads `REG_SIZE` bytes from the address `PC + (1+2) % IDX_MOD` and copies them into `r3`.

```
Terminal
Cycle: 0
Registers:
  ldi(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000000    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 0A A4 00 01 00 02 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 25
Registers:
  ldi(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 01000203    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000007    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 0A A4 00 01 00 02 03 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

`ldi 3, r4, r1` reads `IND_SIZE` bytes from the address `PC + 3 % IDX_MOD`, adds `r4` to this value. `REG_SIZE` bytes are read from the address `PC + SUM % IDX_MOD` and copied into `r1`.

0x0b (sti)

Parameters: `register source, index/register offset_a, index/register offset_b`

Cycles: 25

Behavior:
Adds together `offset_a` and `offset_b` (let's call that SUM) and stores the value of `source` into `PC + SUM % IDX_MOD`

Example:
`sti r2, %4, %5` copies the content of `r2` into the address `PC + (4+5) % IDX_MOD`.

```
Terminal
Cycle: 0
Registers:
  sti(1): alive
    r1 : 00000000    r2 : 00000001    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000000    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 0B 68 02 00 04 00 05 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000020: 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 25
Registers:
  sti(1): alive
    r1 : 00000000    r2 : 00000001    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000007    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 0B 68 02 00 04 00 05 00 00 00 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

`sti r2, %:label, %5`, here we will call the value of `%:label` `label_offset`. It copies the content of `r2` into the address `PC + (label_offset+5) % IDX_MOD`, so 5 bytes after the `label`.

0x0c (fork)

Parameters: `index offset`

Cycles: 800

Behavior:
Creates a new program that inherits different states from the parent.
This program is executed at the address `PC + offset % IDX_MOD`.

Example:
`fork %32` will create a new program, starting at the address `PC + 32 % IDX_MOD`.

```
Terminal
Cycle: 0
Registers:
  fork(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000000    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 0C 00 20 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 800
Registers:
  fork(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000003    carry: 0
   fork(1): alive
     r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
     r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
     r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
     PC : 00000020    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 0C 00 20 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

0x0d (lld)

Parameters: `direct/indirect source, register destination`

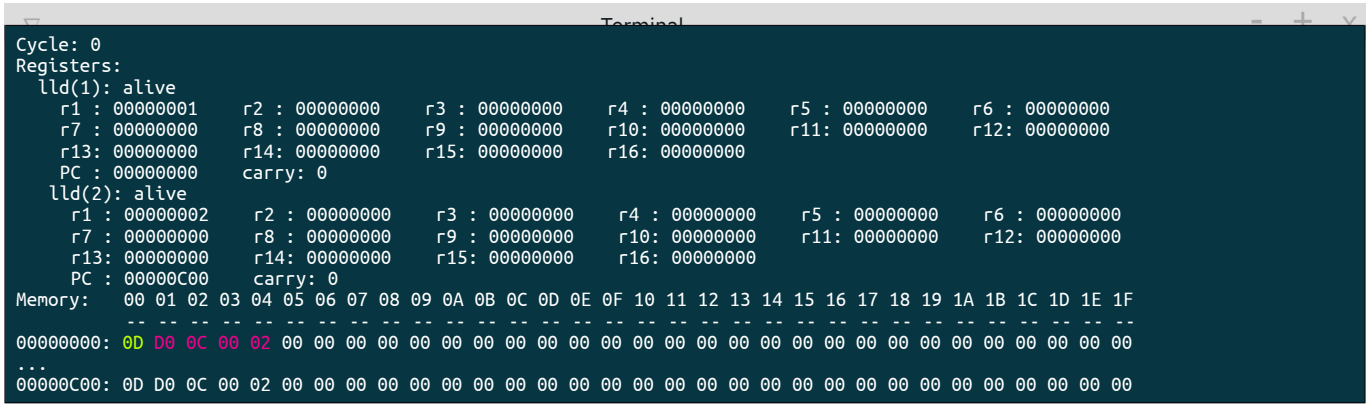
Cycles: 10

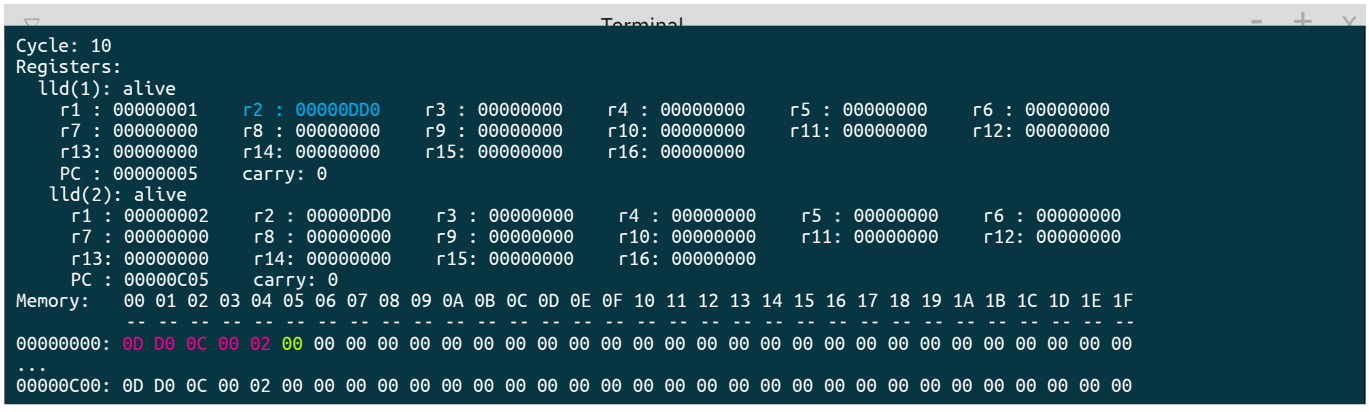
Behavior:
Loads the value of the `source` into `destination`.

This operation sets the carry to 1 if `destination` is zero after the operation, otherwise 0.

Unlike the ***ld*** instruction, ***lld*** does not use the `IDX_MOD` modulo.

Example:
`lld 3072, r2` loads `REG_SIZE` bytes starting at the address `PC + 3072` into `r2`.





0x0e (lldi)

Parameters: `index/register offset_a, index/register offset_b, register destination`

Cycles: 50

Behavior:
Adds together `offset_a` and `offset_b` (let's call that SUM) and loads the value at `PC + SUM` into `destination`. This operation sets the carry to 1 if `destination` is zero after the operation, otherwise 0.

Unlike the ***ldi*** instruction, ***lldi*** does not use the `IDX_MOD` modulo.

Example:
`lldi 2, %3072, r2` reads `IND_SIZ` bytes from the address `PC + 2`, adds 3072 to this value. `REG_SIZE` bytes are read from the address `PC + SUM` and copied into `r2`.

```
Terminal
Cycle: 0
Registers:
  lld(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000000    carry: 0
   lld(2): alive
     r1 : 00000002    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
     r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
     r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
     PC : 00000C00    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 0E E4 00 02 0C 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
...
00000C00: 0E E4 00 02 0C 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 50
Registers:
  lld(1): alive
    r1 : 00000001    r2 : 00020C00    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000005    carry: 0
   lld(2): alive
     r1 : 00000002    r2 : 00020C00    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
     r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
     r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
     PC : 00000C05    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 0E E4 00 02 0C 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
...
00000C00: 0E E4 00 02 0C 00 02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```


0x0f (lfork)

Parameters: `index offset`

Cycles: 1000

Behavior:
Creates a new program that inherits different states from the parent.
This program is executed at the address `PC + offset`.

Unlike the ***fork*** instruction, ***lfork*** does not use the `IDX_MOD` modulo.

Example:
`lfork %3072` will create a new program, starting at the address `PC + 3072 % IDX_MOD`.

```
Terminal
Cycle: 0
Registers:
  lfork(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000000    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 0F 0C 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00000020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

```
Terminal
Cycle: 1000
Registers:
  lfork(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000003    carry: 0
   lfork(1): alive
     r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
     r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
     r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
     PC : 00000020    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 0F 0C 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
...
00000C00: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```


0x10 (print)

Parameters: `register` `character`

Cycles: 2

Behavior:
Displays on the standard output the character whose ASCII code is the content of the register (in base 10). A 256 modulo is applied to this ASCII code.

Example:
`print r3` displays `*` if `r3` contains `42`.

```
Terminal
Cycle: 0
Registers:
  load_and_print_astek(1): alive
    r1 : 00000001    r2 : 00000000    r3 : 00000000    r4 : 00000000    r5 : 00000000    r6 : 00000000
    r7 : 00000000    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000000    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 02 90 00 00 00 41 02 02 90 00 00 00 73 03 02 90 00 00 00 74 04 02 90 00 00 00 65 05 02 90 00 00
00000020: 00 6B 06 02 90 00 00 00 0A 07 10 40 02 10 40 03 10 40 04 10 40 05 10 40 06 10 40 07 00 00 00 00
```

```
Terminal
Cycle: 30
Registers:
  load_and_print_astek(1): alive
    r1 : 00000001    r2 : 00000041    r3 : 00000073    r4 : 00000074    r5 : 00000065    r6 : 0000006B
    r7 : 0000000A    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000000    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 02 90 00 00 00 41 02 02 90 00 00 00 73 03 02 90 00 00 00 74 04 02 90 00 00 00 65 05 02 90 00 00
00000020: 00 6B 06 02 90 00 00 00 0A 07 10 40 02 10 40 03 10 40 04 10 40 05 10 40 06 10 40 07 00 00 00 00
```

```
Terminal
Astek
Cycle: 42
Registers:
  load_and_print_astek(1): alive
    r1 : 00000001    r2 : 00000041    r3 : 00000073    r4 : 00000074    r5 : 00000065    r6 : 0000006B
    r7 : 0000000A    r8 : 00000000    r9 : 00000000    r10: 00000000    r11: 00000000    r12: 00000000
    r13: 00000000    r14: 00000000    r15: 00000000    r16: 00000000
    PC : 00000000    carry: 0
Memory:   00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 1A 1B 1C 1D 1E 1F
          -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
00000000: 02 90 00 00 00 41 02 02 90 00 00 00 73 03 02 90 00 00 00 74 04 02 90 00 00 00 65 05 02 90 00 00
00000020: 00 6B 06 02 90 00 00 00 0A 07 10 40 02 10 40 03 10 40 04 10 40 05 10 40 06 10 40 07 00 00 00 00
```


Are you working on RobotFactory? There are things you might not understand : PC, IDX_MOD, ... Since these are the effects of instructions, they could be interesting elements for Corewar, but maybe not now. Still, you should keep it under your hat in the future, you never know.

v1